INTERNSHIP TASK REPORT

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Task : 01

Basic Network Sniffer

Objective:

This task's goal is to use Python to record and examine network packets.

Tools Used:

- Python
- **Scapy** Library
- Visual Studio Code

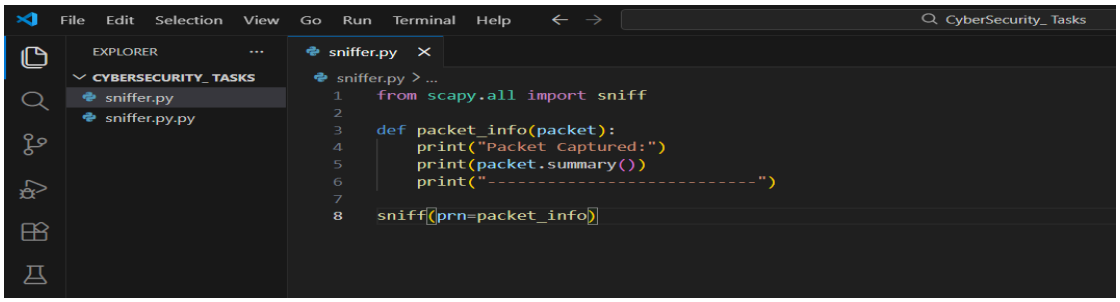
Procedure:

- Installed Python and **Scapy** library.
- Created a Python file named **sniffer.py**.
- Write code to capture network packets.
- Run the program in **VS Code** terminal.
- Observed captured packets in the terminal.

Python Code (sniffer.py) :

```
from scapy.all import sniff
def packet_info(packet):
    print("Packet Captured:")
    print(packet.summary())
    print("---------------------------")
sniff(prn=packet_info)
```

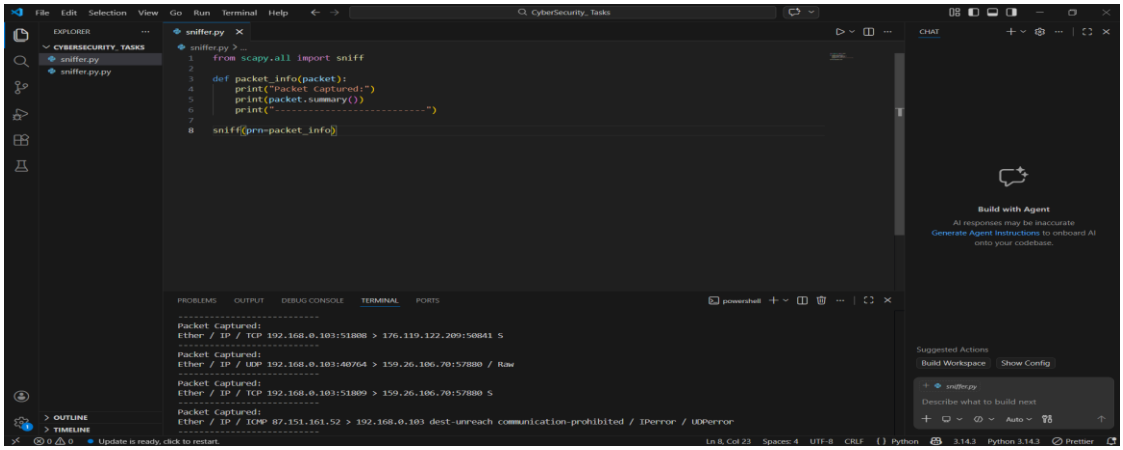
Code Image:



```python
from scapy.all import sniff

def packet_info(packet):
    print("Packet Captured:")
    print(packet.summary())
    print("----------------------------")

sniff(prn=packet_info)
```



Install scapy:

```
PS F:\CyberSecurity_ Tasks> pip install scapy
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: scapy in c:\users\dell\appdata\roaming\python\python314\site-packages (2.7.0)

[notice] A new release of pip is available: 25.3 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
PS F:\CyberSecurity_ Tasks>
```

Output Screenshot's :

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

Ether / IP / UDP 192.168.0.103:40764 > 87.151.160.89:60119 / Raw
----------------------------
Packet Captured:
Ether / IP / UDP 192.168.0.103:40764 > 194.114.136.29:57273 / Raw
----------------------------
Packet Captured:
Ether / IP / UDP 192.168.0.103:40764 > 178.238.10.49:37793 / Raw
----------------------------
Packet Captured:
Ether / IP / UDP 192.168.0.103:40764 > 146.70.171.77:61026 / Raw
----------------------------
Packet Captured:
Ether / IP / UDP 192.168.0.103:40764 > 79.233.202.129:60119 / Raw
----------------------------
```

```
Packet Captured:
Ether / IP / UDP 192.168.0.103:40764 > 89.134.23.215:55388 / Raw
---------------------------
Packet Captured:
Ether / IP / UDP 192.168.0.103:40764 > 89.45.4.9:37594 / Raw
---------------------------
Packet Captured:
Ether / IP / UDP 192.168.0.103:40764 > 66.56.80.80:53990 / Raw
---------------------------
Packet Captured:
Ether / IP / UDP 192.168.0.103:40764 > 46.142.81.232:22418 / Raw
---------------------------
Packet Captured:
Ether / IP / ICMP 46.142.81.66 > 192.168.0.103 dest-unreach communication-prohibited / IPerror / UDPerror
---------------------------
Packet Captured:
Ether / IP / ICMP 46.142.81.232 > 192.168.0.103 dest-unreach communication-prohibited / IPerror / UDPerror
---------------------------
Packet Captured:
Ether / IP / ICMP 66.56.80.80 > 192.168.0.103 dest-unreach port-unreachable / IPerror / UDPerror / Raw
---------------------------
Packet Captured:
Ether / IP / UDP 192.168.0.103:40764 > 79.246.255.140:51413 / Raw
---------------------------
Packet Captured:
Ether / IP / TCP 192.168.0.103:51804 > 79.246.255.140:51413 S
---------------------------
```

Captured Packet's :

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                    powershell

---------------------------
Packet Captured:
Ether / IP / UDP 192.168.0.103:40764 > 176.119.122.209:50841 / Raw
---------------------------
Packet Captured:
Ether / IP / TCP 192.168.0.103:51808 > 176.119.122.209:50841 S
---------------------------
Packet Captured:
Ether / IP / UDP 192.168.0.103:40764 > 159.26.106.70:57880 / Raw
---------------------------
Packet Captured:
Ether / IP / TCP 192.168.0.103:51809 > 159.26.106.70:57880 S
---------------------------
Packet Captured:
Ether / IP / ICMP 87.151.161.52 > 192.168.0.103 dest-unreach communication-prohibited / IPerror / UDPerror
---------------------------
Packet Captured:
Ether / IP / ICMP 87.151.161.52 > 192.168.0.103 dest-unreach communication-prohibited / IPerror / TCPerror
---------------------------
PS F:\CyberSecurity_ Tasks>
```

Result:

The network sniffer's implementation and testing went well. It recorded network packets in real time and showed their summaries, which included protocol details like TCP and UDP.

The task improved comprehension of network traffic analysis and illustrated the use of packet sniffing tools in cybersecurity to track and identify questionable activity.